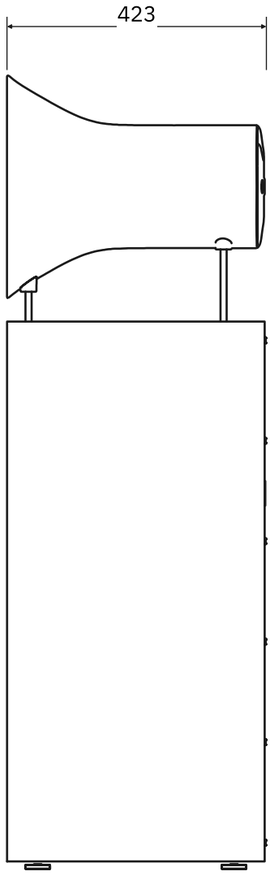
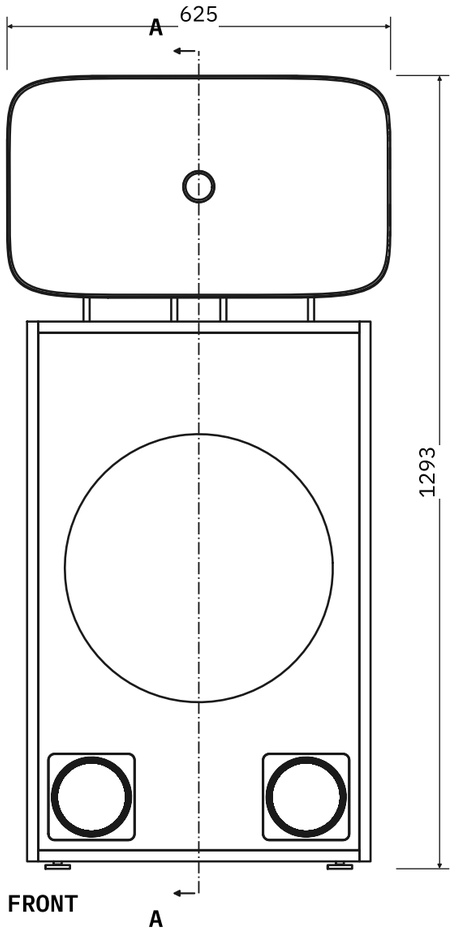
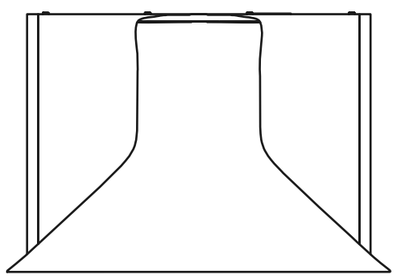
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

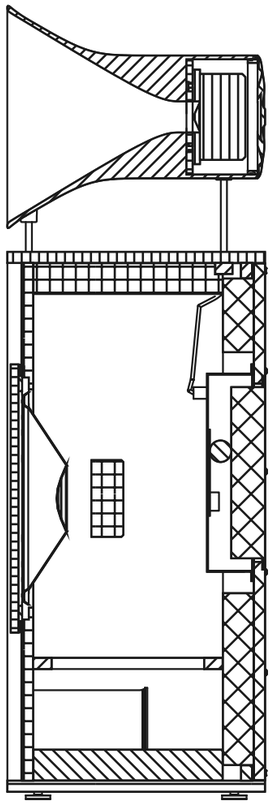
General arrangement

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10. Section A-A (sheet 1): the product cut at x 280.0 and seen from the right; cut faces hatched, a different angle for each part.



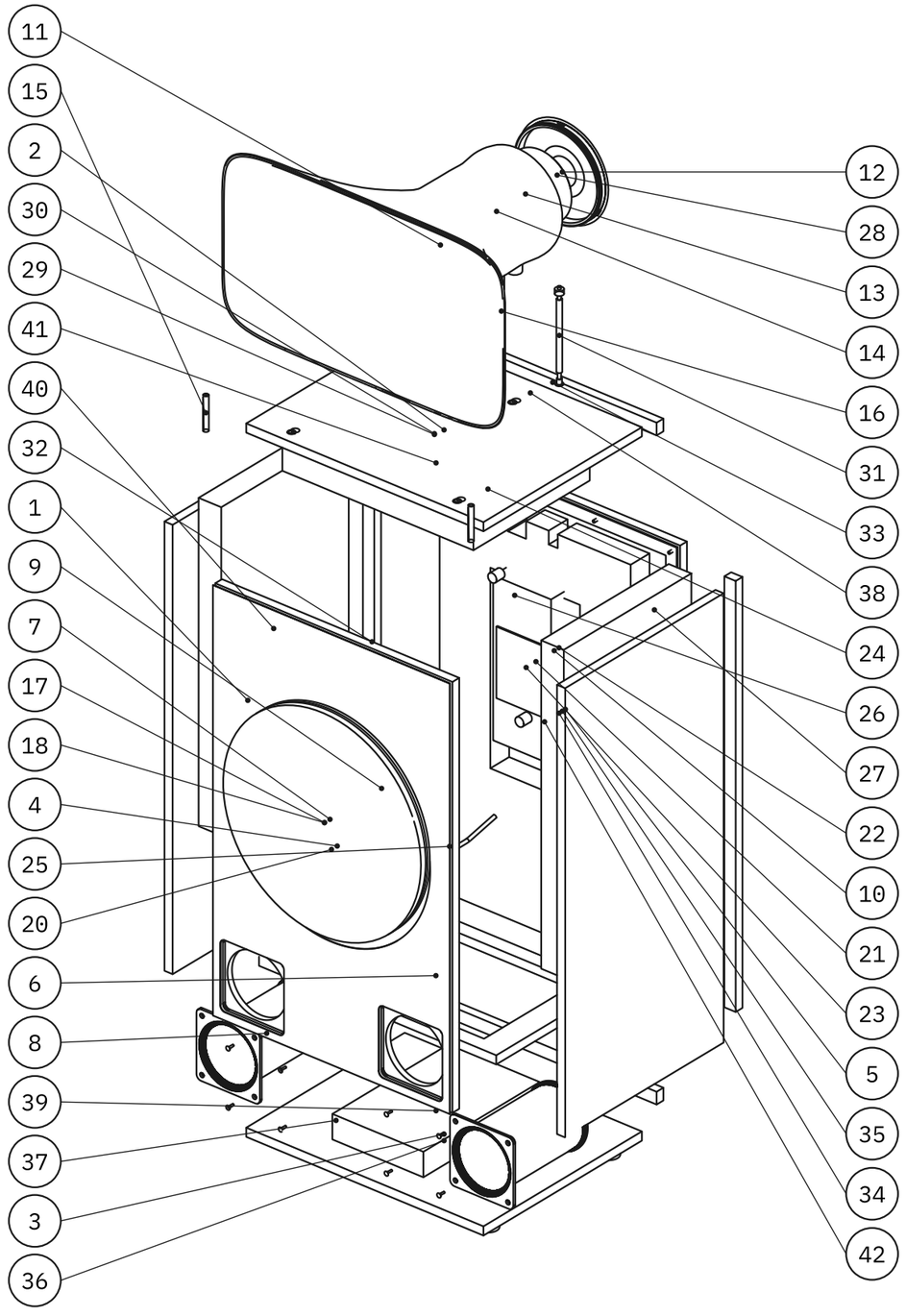
Section A-A

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10, parts drawn apart. Balloons are the parts list's numbers (last sheet).



Exploded view

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-15 | first angle

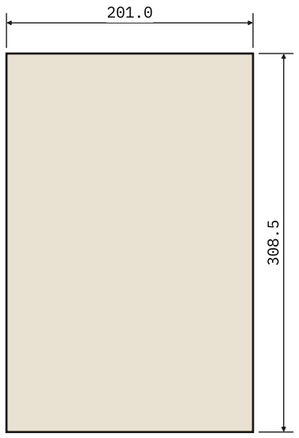
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. Panels cut to their outline only (DXF files in dxf/); sizes as seen from the outer face.



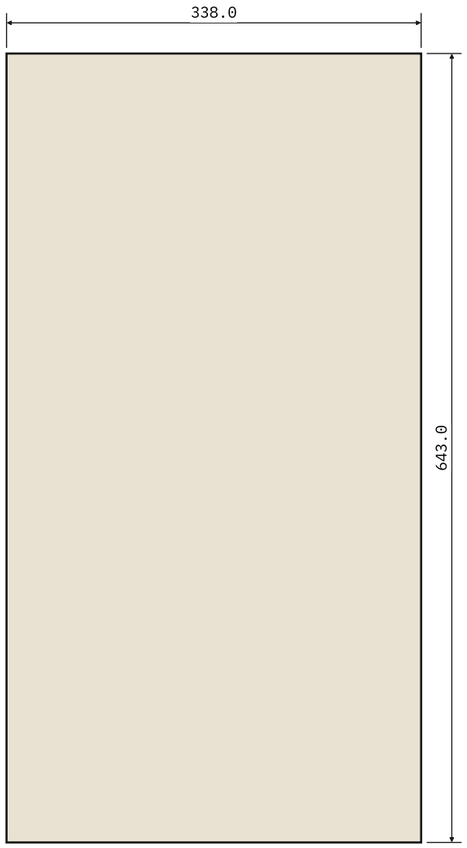
left, right

18 mm baltic-birch, 4 for the set



lining-bottom-1

50 mm polyester-batting, 2 for the set



lining-left-2, lining-right-2

50 mm polyester-batting, 4 for the set

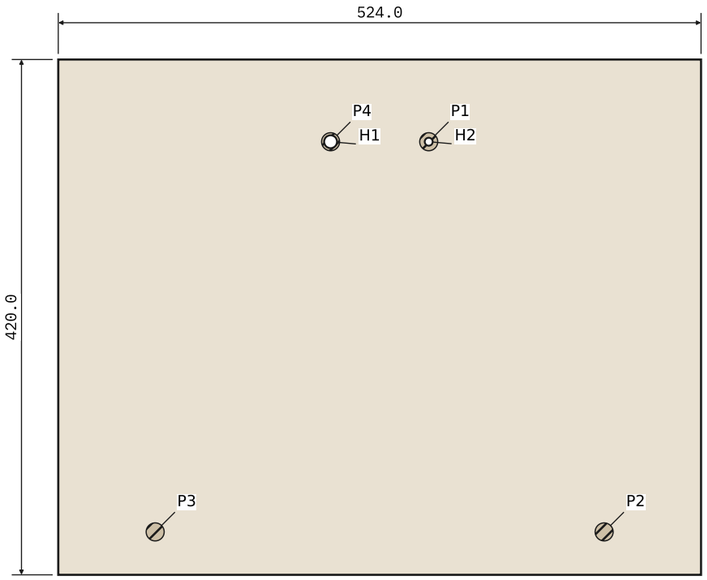
Sheet parts: plain panels

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/top.dxf. Marks listed at the right.



top: 18 mm baltic-birch, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	Ø14.7	302.0	353.0	15 from face
P2	Ø14.7	445.0	35.0	15 from face
P3	Ø14.7	79.0	35.0	15 from face
P4	Ø14.7	222.0	353.0	15 from face
H1	Ø10.5	222.0	353.0	through
H2	Ø6.5	302.0	353.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

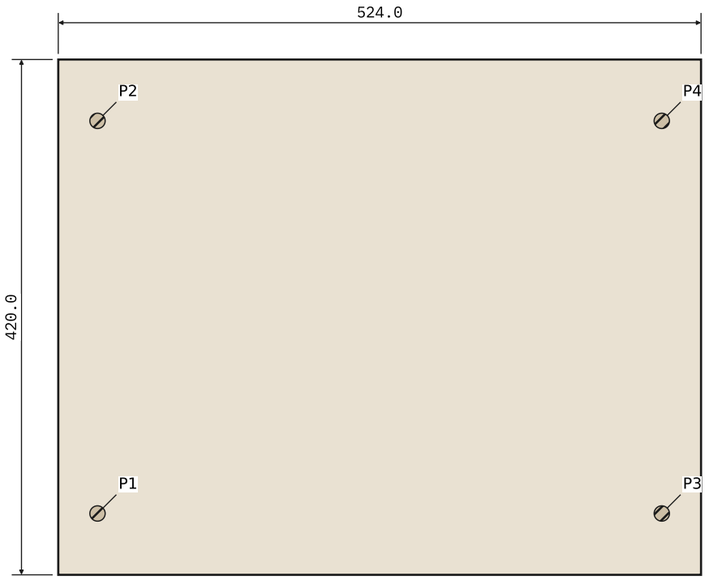
Sheet part: top

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/bottom.dxf. Marks listed at the right.



bottom: 18 mm baltic-birch, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	Ø12.5	32.0	50.0	15 from face
P2	Ø12.5	32.0	370.0	15 from face
P3	Ø12.5	492.0	50.0	15 from face
P4	Ø12.5	492.0	370.0	15 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

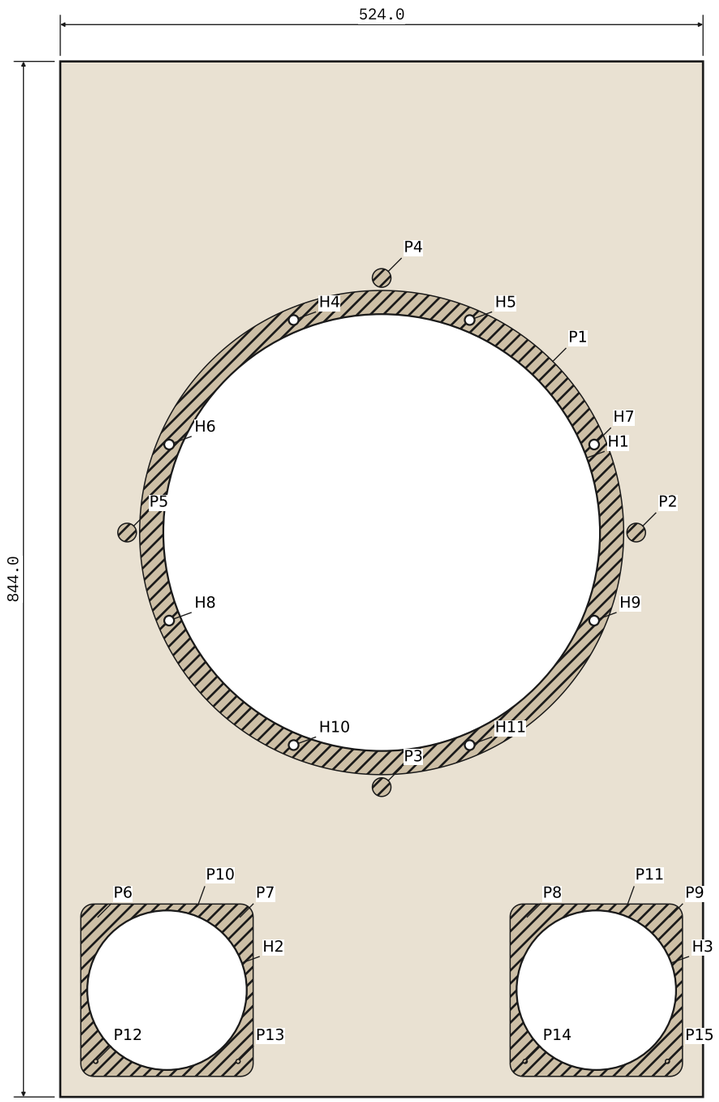
Sheet part: bottom

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/front-panel.dxf. Marks listed at the right.



front-panel: 18 mm baltic-birch, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	Ø394.6	262.0	460.0	11 from face
P2	Ø15.2	469.5	460.0	3.2 from face
P3	Ø15.2	262.0	252.5	3.2 from face
P4	Ø15.2	262.0	667.5	3.2 from face
P5	Ø15.2	54.5	460.0	3.2 from face
P6	profile (DXF)	29.0	145.0	15 from face
P7	profile (DXF)	145.0	145.0	15 from face
P8	profile (DXF)	379.0	145.0	15 from face
P9	profile (DXF)	495.0	145.0	15 from face
P10	profile (DXF)	87.0	87.0	6 from face
P11	profile (DXF)	437.0	87.0	6 from face
P12	profile (DXF)	29.0	29.0	15 from face
P13	profile (DXF)	145.0	29.0	15 from face
P14	profile (DXF)	379.0	29.0	15 from face
P15	profile (DXF)	495.0	29.0	15 from face
H1	Ø356.0	262.0	460.0	through
H2	Ø130.0	87.0	87.0	through
H3	Ø130.0	437.0	87.0	through
H4	Ø8.0	190.2	633.2	through
H5	Ø8.0	333.8	633.2	through
H6	Ø8.0	88.8	531.8	through
H7	Ø8.0	435.2	531.8	through
H8	Ø8.0	88.8	388.2	through
H9	Ø8.0	435.2	388.2	through
H10	Ø8.0	190.2	286.8	through
H11	Ø8.0	333.8	286.8	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

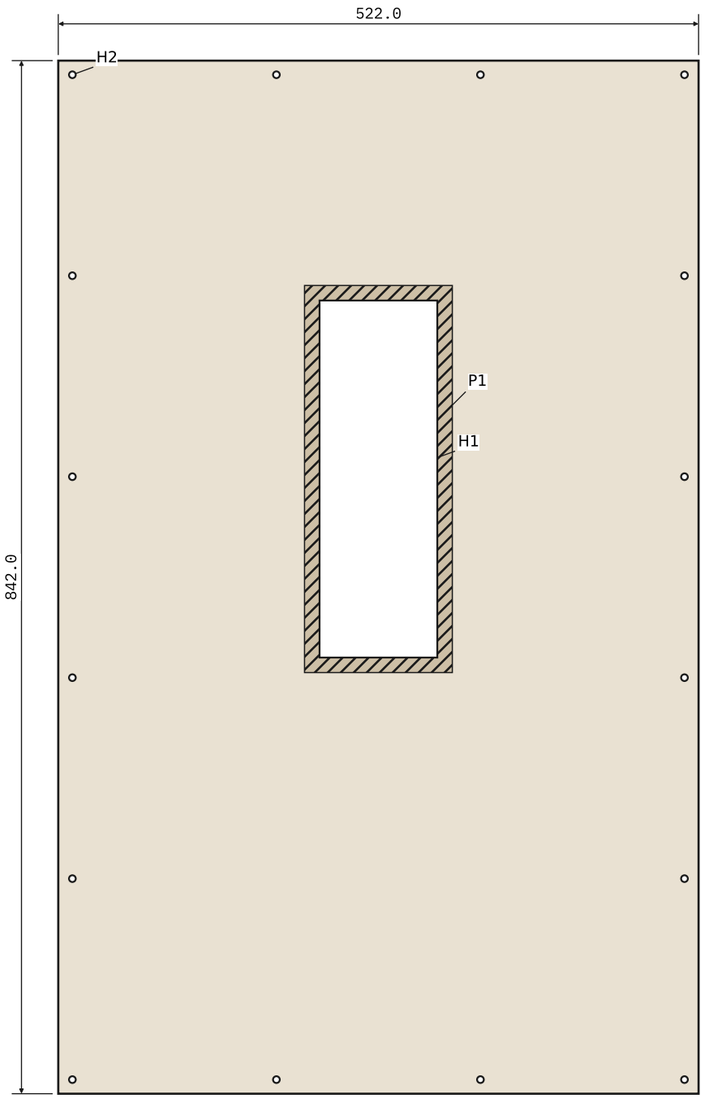
Sheet part: front-panel

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/back.dxf. Marks listed at the right.



back: 18 mm baltic-birch, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	120.5 x 315.5	261.0	501.0	4.5 from face
H1	96.0 x 291.0	261.0	501.0	through
H2	Ø5.5	16 off	DXF	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: back

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

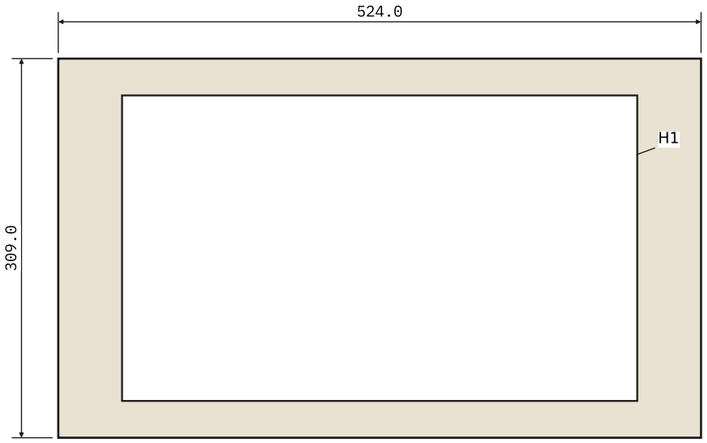
mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/brace.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	420.0 x 249.0	262.0	154.5	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



brace: 18 mm baltic-birch, 2 for the set, seen from its outer face

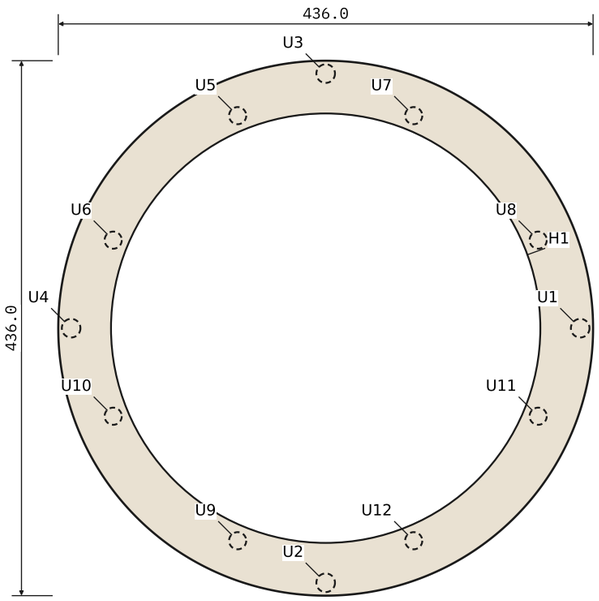
Sheet part: brace

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/woofer-ring.dxf. Marks listed at the right.



woofer-ring: 6 mm mdf, 2 for the set, seen from its outer face

inner-face pockets: dxf/woofer-ring-underside.dxf (seen from that face)

mark	size	x	y	depth
H1	Ø350.0	218.0	218.0	through
U1	Ø15.2	425.5	218.0	3.2 from inside
U2	Ø15.2	218.0	10.5	3.2 from inside
U3	Ø15.2	218.0	425.5	3.2 from inside
U4	Ø15.2	10.5	218.0	3.2 from inside
U5	Ø14.0	146.3	391.2	4 from inside
U6	Ø14.0	44.8	289.8	4 from inside
U7	Ø14.0	289.8	391.2	4 from inside
U8	Ø14.0	391.2	289.8	4 from inside
U9	Ø14.0	146.3	44.8	4 from inside
U10	Ø14.0	44.8	146.2	4 from inside
U11	Ø14.0	391.2	146.2	4 from inside
U12	Ø14.0	289.8	44.8	4 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

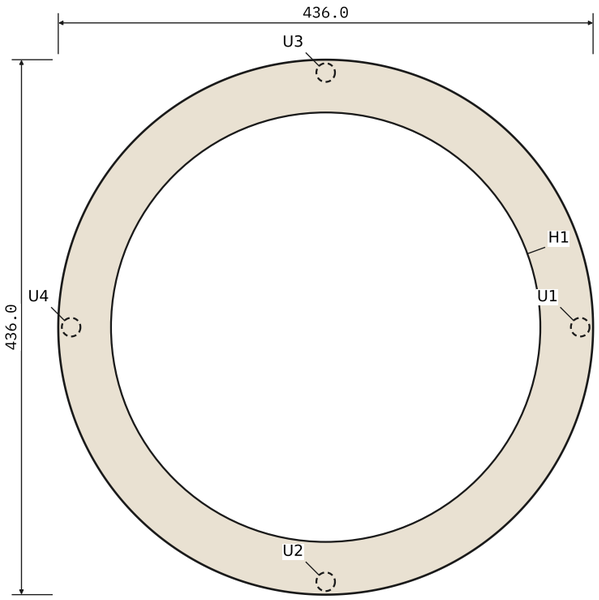
Sheet part: woofer-ring

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/disc-frame.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	Ø350.0	218.0	218.0	through
U1	Ø15.2	425.5	218.0	3.2 from inside
U2	Ø15.2	218.0	10.5	3.2 from inside
U3	Ø15.2	218.0	425.5	3.2 from inside
U4	Ø15.2	10.5	218.0	3.2 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

disc-frame: 12 mm mdf, 2 for the set, seen from its outer face

inner-face pockets: dxf/disc-frame-underside.dxf (seen from that face)

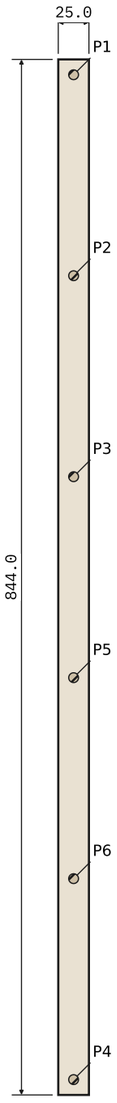
Sheet part: disc-frame

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/land-left.dxf. Marks listed at the right.



mark	size	x	y	depth
P1	Ø8.0	12.5	831.5	12 from face
P2	Ø8.0	12.5	667.7	12 from face
P3	Ø8.0	12.5	503.9	12 from face
P4	Ø8.0	12.5	12.5	12 from face
P5	Ø8.0	12.5	340.1	12 from face
P6	Ø8.0	12.5	176.3	12 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

land-left: 18 mm baltic-birch, 2 for the set, seen from its outer face

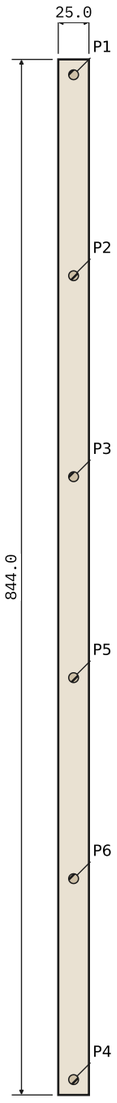
Sheet part: land-left

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/land-right.dxf. Marks listed at the right.



mark	size	x	y	depth
P1	Ø8.0	12.5	831.5	12 from face
P2	Ø8.0	12.5	667.7	12 from face
P3	Ø8.0	12.5	503.9	12 from face
P4	Ø8.0	12.5	12.5	12 from face
P5	Ø8.0	12.5	340.1	12 from face
P6	Ø8.0	12.5	176.3	12 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

land-right: 18 mm baltic-birch, 2 for the set, seen from its outer face

Sheet part: land-right

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

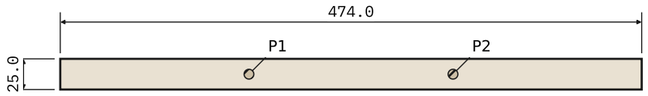
mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/land-top.dxf. Marks listed at the right.

mark	size	x	y	depth
P1	Ø8.0	153.8	12.5	12 from face
P2	Ø8.0	320.2	12.5	12 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



land-top: 18 mm baltic-birch, 2 for the set, seen from its outer face

Sheet part: land-top

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

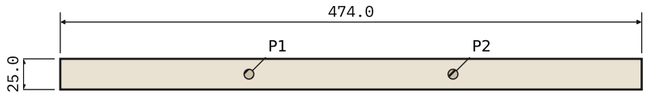
mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/land-bottom.dxf. Marks listed at the right.

mark	size	x	y	depth
P1	Ø8.0	153.8	12.5	12 from face
P2	Ø8.0	320.2	12.5	12 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



land-bottom: 18 mm baltic-birch, 2 for the set, seen from its outer face

Sheet part: land-bottom

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

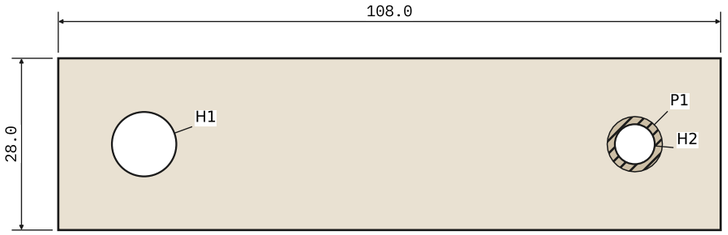
mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/top-pad.dxf. Marks listed at the right.

mark	size	x	y	depth
P1	Ø9.0	94.0	14.0	10 from face
H1	Ø10.5	14.0	14.0	through
H2	Ø6.5	94.0	14.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



top-pad: 18 mm baltic-birch, 2 for the set, seen from its outer face

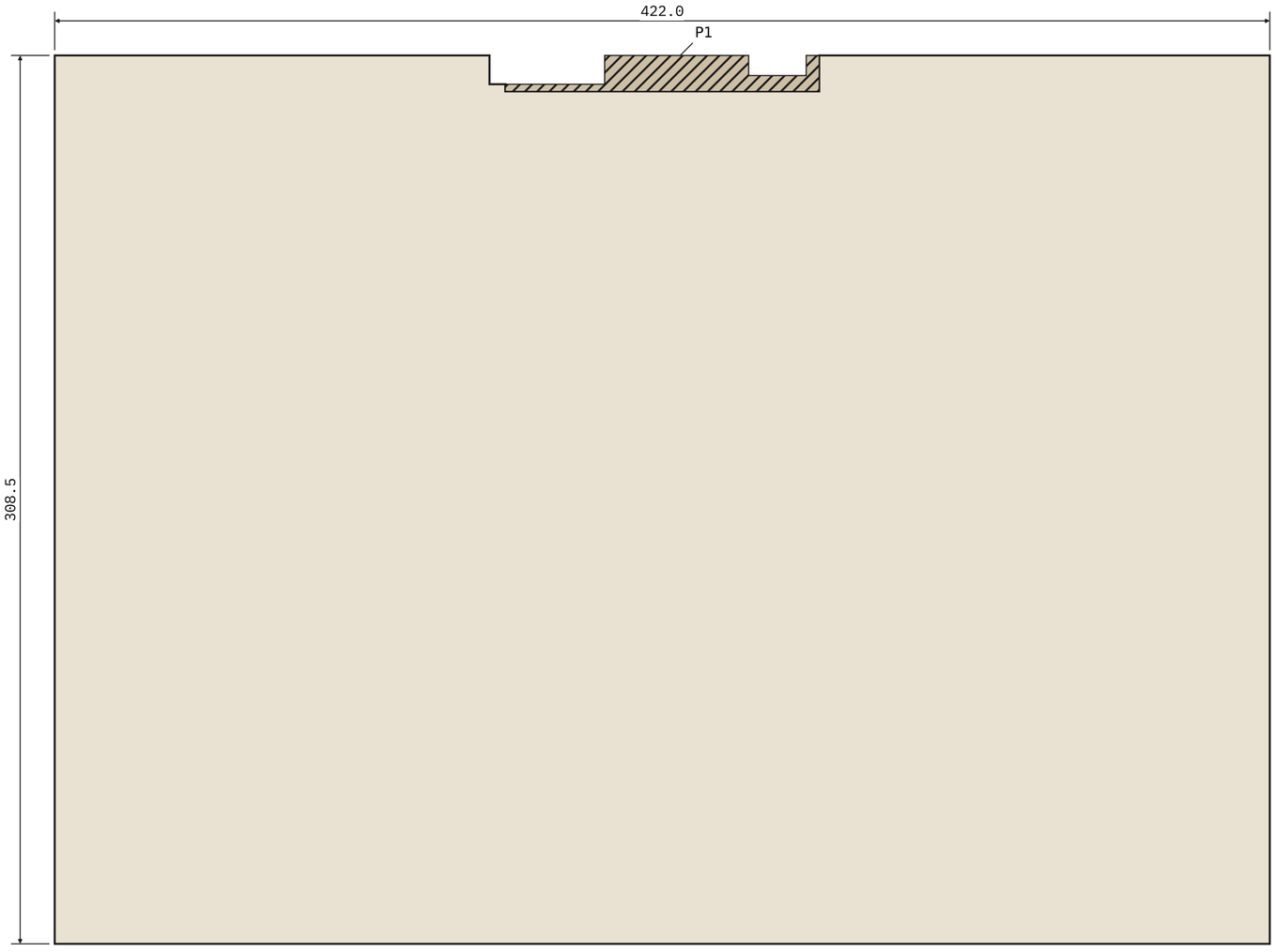
Sheet part: top-pad

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/lining-top-2.dxf. Marks listed at the right.



mark	size	x	y	depth
P1	profile (DXF)	211.0	302.2	18 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

lining-top-2: 50 mm polyester-batting, 2 for the set, seen from its outer face

Sheet part: lining-top-2

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

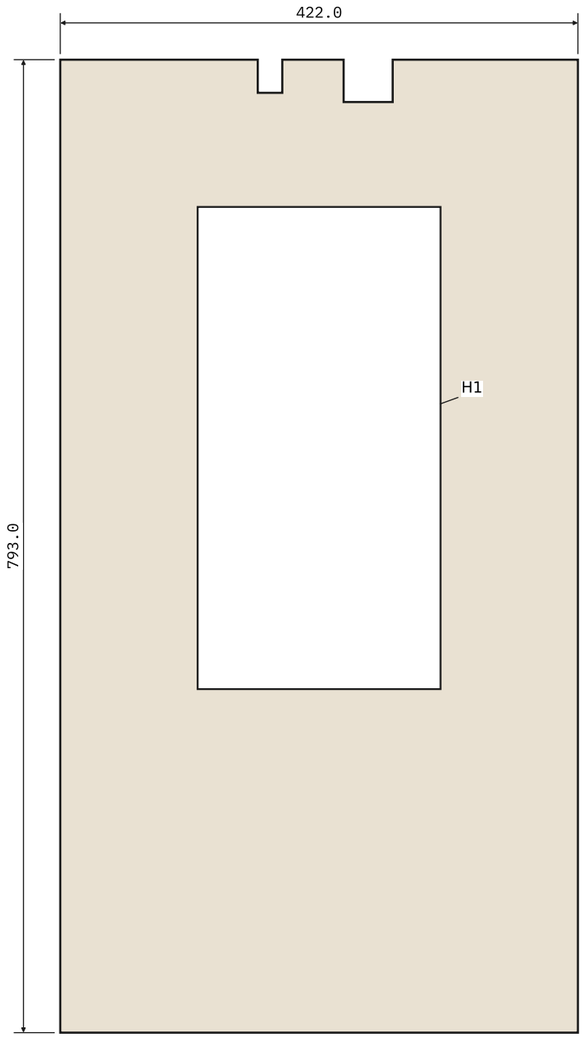
mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

17 of 26

BAR-017 rev A

Scale 1:5. As the router sees it from the outer face; DXF: dxf/lining-back.dxf. Marks listed at the right.



lining-back: 50 mm polyester-batting, 2 for the set, seen from its outer face

mark	size	x	y	depth
H1	198.0 x 393.0	211.0	476.5	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: lining-back

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

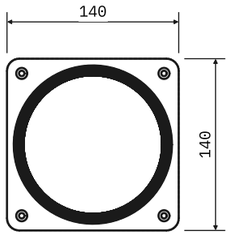
mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

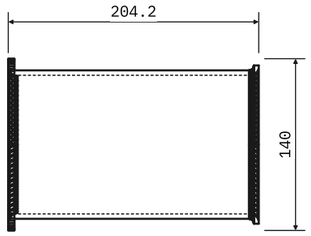
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

8. port-left, port-right: asa print, on its frame; 337 cm3 each; 4 for the set; 1:5

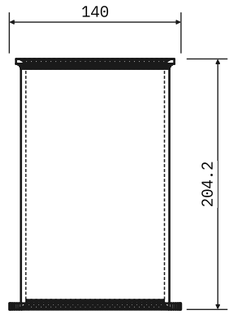
113 across inside, 204 long from the panel's face, its 140 square frame (corners R10) printed in one, both mouths rounded on R8: the 2 together tune the box to 38 Hz; its length found on the first sample (three made 10 apart, the one whose impedance minimum falls at fb kept) and the rest made to it: its rounded ends cannot be trimmed; fitted from the front, the frame flush in a 6 recess on a 1 closed-cell PE foam gasket, four black countersunk wood screws in its corners



FRONT



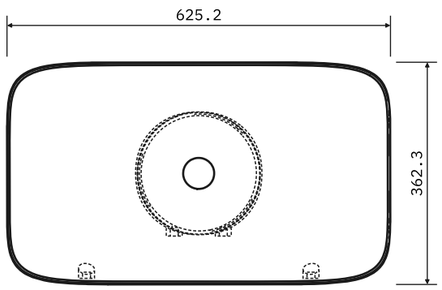
FROM THE RIGHT



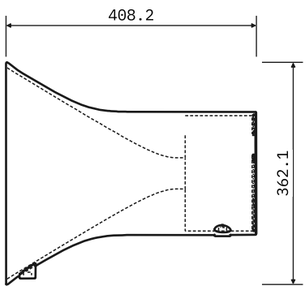
FROM ABOVE

11. waveguide: cast, cast-resin; 5218 cm3 each; 2 for the set; 1:10

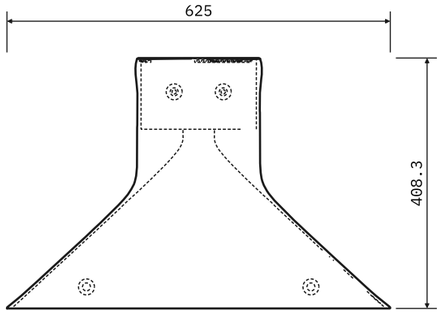
mouth 610 x 346, 278 deep, 2 in throat; a 10 rim rounded right round, 625 x 362 outside; a shell 6 thick following its flare; the driver in a pod 200 across behind, closed by a cap; four pads under it take the posts' brass inserts; cast in tinted clear resin in a two-part silicone mould from a CNC-cut plug, its faces polished satin (a slow deep-pour resin: the neck round the throat is solid, 5.2 L in all), or pigmented the same way for the opaque and marbled finishes



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

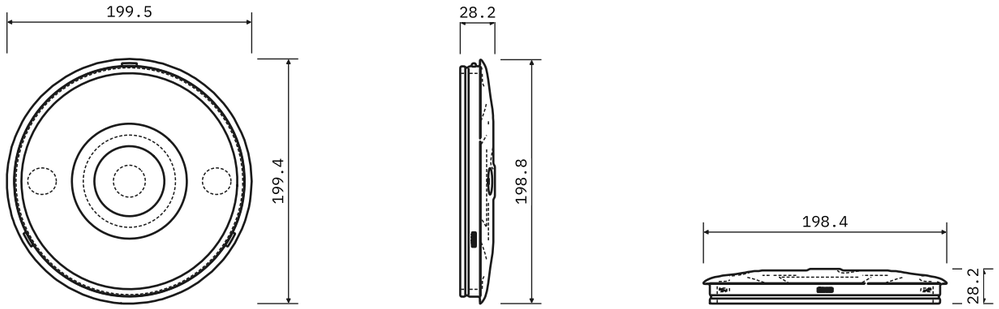
mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

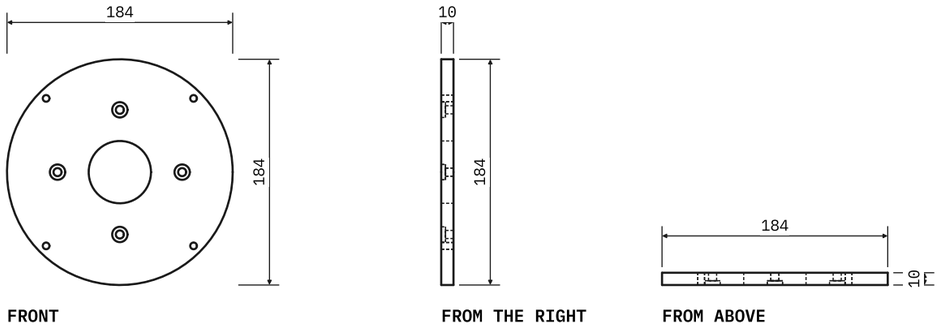
12. waveguide-cap: cast, cast-resin; 204 cm3 each; 2 for the set; 1:5

closes the pod behind the driver on a twist-lock: three lugs cast on its spigot go into L-slots in the pod's bore and an eighth of a turn (45 degrees, clockwise from behind) locks them, a small bump in each slot clicking them home; an O-ring on the spigot, beyond the lugs, seals it and stops it rattling; two shallow finger dimples to turn it by; no screw shows; the HF pair leaves through a post below; unplugged here, the horn lifts off its sockets



14. hf-adaptor: lathe, aluminium; 242 cm3 each; 2 for the set; 1:5

184 across, 10 thick, bored 50.8 (the throat); 4 countersunk M6 on 101.6 take the driver from the front; 4 x 5.5 on 170 (at 45 degrees to them) take M5 socket caps into the brass inserts in the horn's boss; black anodised



Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

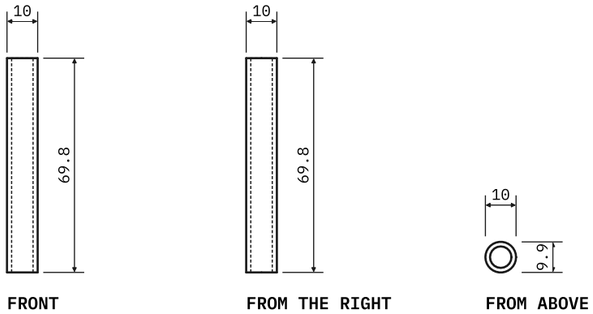
mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

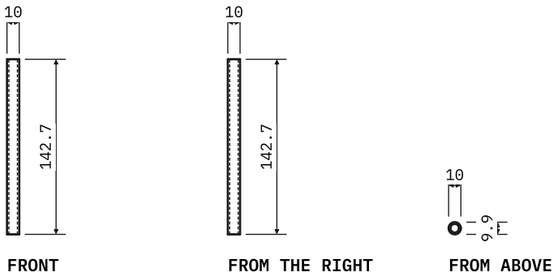
15. post-1, post-2: lathe, stainless; 3 cm3 each; 4 for the set; 1:2

70 long (49 showing): stainless tube 10 x 7, threaded M10 x 1 8 at the top into the brass insert in its pad; its foot plain, standing 13 deep in a polyurethane sleeve in a brass socket in the top; its bore deburred



16. post-3, post-4: lathe, stainless; 6 cm3 each; 4 for the set; 1:5

143 long (118 showing): stainless tube 10 x 7, threaded M10 x 1 12 at the top into the brass insert in its pad; its foot plain, standing 13 deep in a polyurethane sleeve in a brass socket in the top; its bore deburred; the HF lead runs down inside it



Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

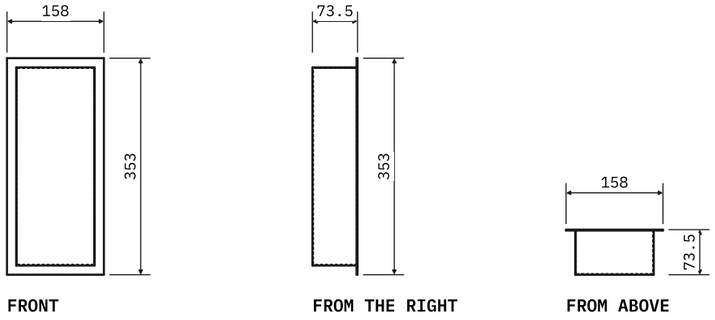
mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

21. amp-box: sheet metal, aluminium; 180 cm3 each; 2 for the set; 1:10

125 x 320 inside, 75 deep from the back: 1.5 aluminium folded, its corners welded, black anodised; a 15 flange round its open side, on a 3 EPDM gasket on the back's inner face, ten screws through it, so the box stays airtight and the mains electronics are enclosed in metal; two M12 cable glands in its front; the ports' length found with it fitted (it takes volume)



Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 2; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	left, right	cut	4	18 mm Baltic birch plywood, BB/BB	420 x 880	26.80
2	top	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 420	15.96
3	bottom	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 420	15.96
4	front-panel	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 844	32.06
5	back	cut	2	18 mm Baltic birch plywood, BB/BB	522 x 842	31.87
6	brace	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 309	11.74
7	woofer-ring	cut	2	6 mm MDF	436 x 436	4.28
8	port-left, port-right	print	4	ASA, FDM (outdoor, paintable)	337 cm3	31.61
9	woofer	buy	2	FaitalPRO 15PR400-8 15 in woofer (treated paper cone, neodymium, 8 ohm)	393 frame, 356 cut-out, 165 deep	194.00
10	amp-plate	buy	2	Hypex FusionAmp FA122 DSP plate amplifier, 2 x 125 W into 4 ohm	315 x 120 x 55 plate, 291 x 96 recess	422.40
11	waveguide	cast	2	Casting resin (polyurethane or epoxy), tinted clear, translucent or pigmented	5218 cm3, 635 x 418 x 372	684.59
12	waveguide-cap	cast	2	Casting resin (polyurethane or epoxy), tinted clear, translucent or pigmented	204 cm3, 209 x 38 x 209	138.97
13	hf	buy	2	B&C Speakers DE1085TN 2 in compression driver, titanium diaphragm, 100 mm voice coil, neodymium, 16 ohm (round 5: the owner's bigger throat)	154 across, 86 deep, 4 x M6 on 102	512.00
14	hf-adaptor	lathe	2	Aluminium 6061	194 x 20 x 194	43.82
15	post-1, post-2	lathe	4	Stainless steel 304	20 x 20 x 80	15.96
16	post-3, post-4	lathe	4	Stainless steel 304	20 x 20 x 153	16.83
17	disc-frame	cut	2	12 mm MDF	436 x 436	4.28
18	cloth-disc	buy	2	Acoustically transparent speaker cloth (a knit sold for grilles), about 0.6 x 0.95 m a grille, in the finish's colour		15.00
19	cloth-disc-magnets	buy	2	The ring's and the disc's magnets: eight 15 x 3 neodymium discs (N42) and four 15 x 3 mild-steel discs, a set: four magnets in the panel and the steel discs in the ring's back hold the ring, four in the disc's frame hold the disc to the same		10.00
20	bark-face, bark	buy	6	Bark for laminating, a square metre: paper birch (Betula papyrifera) sheet from fallen or felled trees, or a smooth-barked gum's shed bark pressed flat (or a UV-printed veneer of it); approve a sample board first		[PRICE]
21	amp-box	sheet metal	2	Aluminium 6061	305 x 500 x 2	85.72
22	amp-box-gasket	buy	2	EPDM sponge gasket, 3 mm, cut as a frame 15 wide for the amplifier compartment's flange		6.00
23	hf-pad	buy	2	The HF channel's L-pad: two aluminium-housed wirewound resistors, 25 W (Vishay Dale RH-25 or Arcol HS25 kind), 10 ohm in series and 8.2 ohm across the driver (about -9 dB at 16 ohm; trimmed after the first measurement), screwed with thermal		24.00
24	lead-hf	buy	2	HF lead: round silicone-sheathed cable 2 x 0.5 mm2, about 5 across, 1.3 m, ferrules at the pod's end and a two-way lever connector (Wago 221-412 or like) on the driver's own leads, a polarised 2-way connector at the compartment's end		6.00
25	lead-woofer	buy	2	Round-sheathed speaker cable 2 x 1.5 mm2, 1 m		2.00
26	cable-glands	buy	2	Three M12 nylon cable glands for 3 to 6.5 mm cable, black: two in the amplifier compartment's front, one under the top at the cable post, in an M12 insert for plastics (in acrylic, a thread tapped in the top)		4.00

Parts list (1 of 3)

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 2; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
27	rating-plate	buy	2	Rating plate 70 x 40 x 1, black anodised aluminium, laser-marked: the model, the serial number, the ratings, the marks (CE, UKCA, FCC, the WEEE bin) and the maker's address; two M3 screws into inserts		10.00
28	cap-o-ring	buy	2	O-ring for the horn's cap, 182 x 2 mm, EPDM 70 Shore A (round 16: the twist-lock), and a tube of silicone grease		4.00
29	post-sockets	buy	2	The posts' brass fittings: four M10 x 1 threaded inserts bonded with epoxy into the waveguide's pads after casting, in holes drilled through a jig taken off the top's sockets (13 across; the front pair 10 long, the rear 14, the tie post's		48.00
30	post-sleeves	buy	2	Polyurethane sleeves 12 x 10 x 13, Shore A 90, a set of four: pressed into the posts' sockets, they centre the posts and stop them buzzing		8.00
31	tie-rod	buy	2	The horn's tie-down: an M6 A2 threaded rod about 110 long, an M6 knurled thumb nut and a washer, and a flanged M6 threaded insert (body 9 x 10) for the top's underside: one for plastics in HDPE, bonded with epoxy in acrylic		8.00
32	land-left, land-right	cut	4	18 mm Baltic birch plywood, BB/BB	25 x 844	1.53
33	land-top, land-bottom	cut	4	18 mm Baltic birch plywood, BB/BB	474 x 25	0.86
34	back-gasket	buy	2	EPDM sponge strip 25 x 3, closed cell, cut and joined as a frame on the back's four land strips, punched for its 16 screws		8.00
35	back-screws	buy	2	16 M5 button heads (ISO 7380, A2 stainless, black: round 16) on black nylon washers: the back to the inserts in its land strips (M5 x 40 into HDPE or acrylic, M5 x 30 into plywood)		8.00
36	feet	buy	2	Levelling feet M8, a 40 mm swivel base on a rubber pad, black, with lock nuts, a set of 4		18.00
37	port-screws	buy	2	The ports' fixings: eight black countersunk screws, four through each port's frame into the panel (thread-forming screws for plastics, EJOT PT K40 kind, 4 x 16 in the 20 mm HDPE; M4 x 20 into tapped threads in the 25 mm acrylic; wood screws 4 x		5.00
38	top-pad	cut	2	18 mm Baltic birch plywood, BB/BB	108 x 28	0.22
39	lining-bottom-1	cut	2	50 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	201 x 308	0.78
40	lining-left-2, lining-right-2	cut	4	50 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	338 x 643	2.72
41	lining-top-2	cut	2	50 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	422 x 308	1.63
42	lining-back	cut	2	50 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	422 x 793	4.18
43	woofer-fixings	buy	2	M6 x 25 button heads (A2) with M6 T-nuts, a set of 8 (the 15PR400's eight holes on 375)		8.00
44	hf-fixings	buy	2	The 2 in driver's fixings: four M6 x 16 countersunk screws (A2), the driver to its adaptor ring, and four M5 x 16 socket caps with washers, the ring to the brass inserts in the horn's boss, a set		6.00
45	gasket	buy	4	1.5 mm closed-cell foam gasket strip, 1 m		5.00

Parts list (2 of 3)

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 2; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
46	lining-sleeves	buy	2	White polyester voile (a sheer curtain fabric), about 3 m2 a speaker with its seams, sewn into a sleeve round each batt of the lining, so no fibre reaches the ports or the woofer and the batt's face stays even		10.00
47	amp-fixings	buy	2	Hypex 4.3 x 16 self-tapping pan heads and 3 mm EPDM tape, a plate's set		10.00
48	amp-box-fixings	buy	2	Ten screws for the amplifier compartment's flange, with washers: thread-forming screws for plastics (EJOT PT K40 x 12 kind) into HDPE, M4 screws into inserts bonded in acrylic, or wood screws into plywood		3.00
49	hf-capacitor	buy	2	33 uF 100 V polypropylene film capacitor, in series ahead of the L-pad (a first-order high-pass at about 300 Hz into 16 ohm)		12.00
50	loom	buy	2	The loom: screw-mount cable-tie bases with thread-forming screws for plastics (EJOT PT K30 x 10 kind) every 120, ties, a polarised 2-way connector (Molex Mini-Fit Jr or Wago 2092 kind) for each lead at the compartment, crimped insulated 6.3		18.00
51	terminals	buy	8	6.3 mm insulated push-on terminal		0.50
52	back-inserts	buy	2	M5 threaded inserts for wood (hex drive), a set of 16, in 8 holes 12 deep in the back's plywood cleats		8.00
53	feet-inserts	buy	2	M8 threaded inserts for wood (hex drive), a set of 4, in 12.5 holes 15 deep in the plywood bottom, for the feet		4.00

Parts list (3 of 3)

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Notes

1. Bark on the shell (sides, top, bottom and back): real birch bark sheet (Betula papyrifera, from fallen or felled trees) laminated to the plywood with PVA in a vacuum bag and sealed with a matt waterborne lacquer, wrapped round the shell's front edges; for eucalyptus, the shed bark of a smooth-barked gum (spotted or snow gum) pressed flat and laminated the same way, or a UV-printed veneer of it. Approve a sample board first.
2. Glue-up: sides to the bottom, the brace, then the front panel (set back 24 mm), the top, and the back's four cleats, their back faces 21 in from the shell's back edge; the back screws on last, on its gasket, and comes off.
3. The woofer flush in its rebate on a gasket, on eight M6 T-nuts from inside; the ring over its frame, held by four magnets in the panel under steel discs in its back; each port, printed in one with its square frame, fitted from the front, the frame flush in a recess on a foam gasket, four black wood screws in its corners; its length found on the first sample (its rounded ends cannot be trimmed).
4. The waveguide stands on four posts, stainless lamp tube 10 x 7, each threaded M10 x 1 at its top into a brass insert bonded in a pad cast under the horn (drilled after casting, through a jig taken off the top's sockets), its plain foot standing in a polyurethane sleeve in a brass socket pressed into the top, so the horn and its posts lift off as one. An M6 rod down the free rear post, from a thumb nut inside the pod to an insert under the top, holds them down. The cap closing the pod is held by a twist-lock (round 16): three lugs on its spigot in L-slots in the pod's bore, locked by an eighth of a turn, an O-ring beyond them; no screw shows. The HF lead, a round silicone cable 5 across, runs from its connector at the amplifier's compartment straight up, through a gland under the top and up inside a rear post into the pod, where it plugs into the driver's leads. To lift the horn off: the cap off, the nut undone and the lead unplugged.
5. The front: the panel faced in bark (1 mm) round the ring and the port, and a cloth disc over the woofer, its MDF frame standing on the ring on four magnet pairs, its face 18.5 clear of the cone at rest.
6. The FA122 crosses at about 800 Hz (LR4: B&C asks 12 dB/oct or steeper there) with the HF channel about 9 dB down (108 dB at 2.83 V against the woofer's 99), part of it in the passive L-pad after the amplifier, so its hiss stays inaudible: two 25 W aluminium-housed resistors on a plate in the compartment, behind a 33 uF film capacitor that keeps low frequencies off the driver if the preset is ever lost.
7. The amplifier's preset, stored before the speaker ships: the crossover; a 4th-order Butterworth high-pass at 28 Hz on the woofer (below the ports' tuning the box no longer holds the cone, which would pass Xmax below about 30 Hz at full output, and a record's warp drives it there); limiters set to the woofer's Xmax and the driver's rating; a notch at the ports' pipe resonance (about 540 Hz in the pressed cabinet, 560 in the Cloud's, 600 in the plywood one), measured in each port's near field.
8. The amplifier in a sealed compartment of its own, as Hypex asks: 1.5 mm aluminium, black anodised, on an EPDM gasket on the back's inner face, so the box stays airtight and the mains electronics are enclosed in metal. The leads leave it through cable glands. A rating plate beside the amplifier carries the model, the serial number, the ratings, the marks and the maker's address; the finished speaker needs an EMC pre-scan and an IEC 62368-1 assessment.
9. The lining: polyester batts (thermally bonded, recycled PET, 50 mm) on the sides, the top and the bottom, one to each bay between the braces (and the ribs), and on the back, its full height between its ribs, so they come off with it; each cut 10 full each way and sewn into a sleeve of white voile; none on the front panel, none beside the ports (10 clear of each tube, nothing between its inner end and the back's batts) and none within 20 of the amplifier's compartment.
10. The back comes off (round 14): it closes on a 3 mm EPDM gasket against four land strips inside the shell, sixteen M5 button heads through it into inserts in them. The amplifier's compartment, the back's ribs and its batts come off with it and the leads unplug at the compartment, so the amplifier, the L-pad and the leads can be reached.
11. Four levelling feet (M8, 40 swivel bases on rubber pads) in inserts in the bottom, 50 in from its edges, lift the cabinet 12 mm: set them so it stands level and cannot rock under the horn, and lock their nuts.

Notes

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15